



VGH-TP VPC workflow

Jason

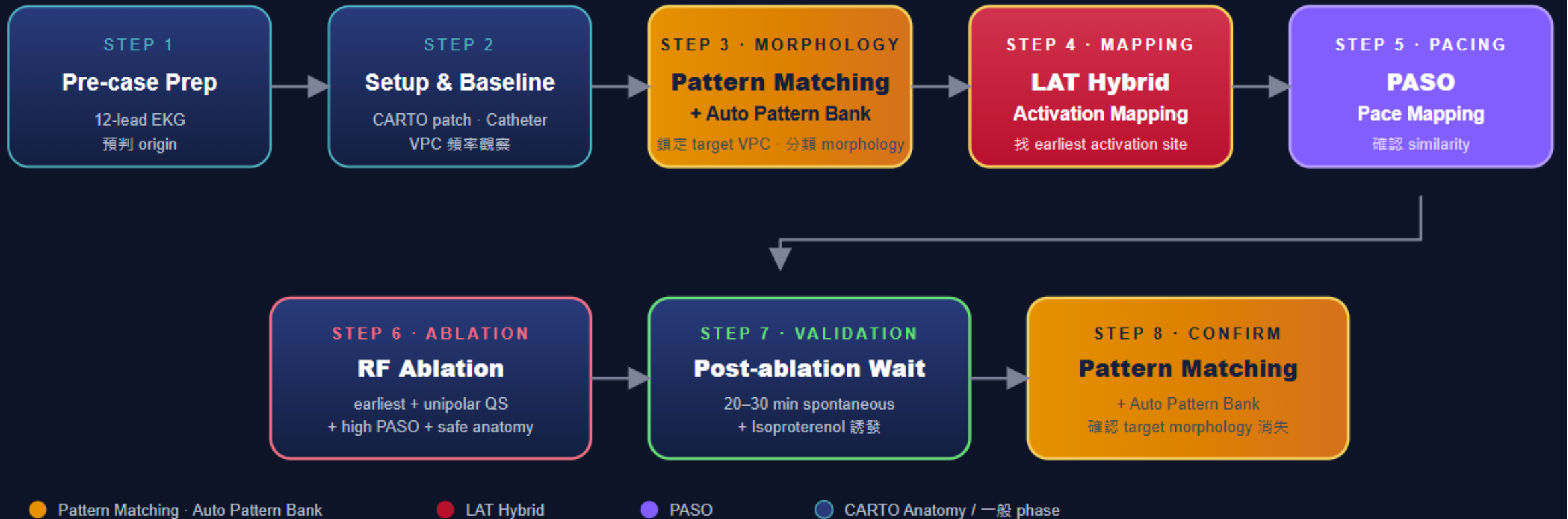


Step 1 Pre-case Prep

- 備機前確認
 - 導管室
 - 主診斷 (AF, VPC, PSVT.....)
 - 主治醫師 (手術習慣、常用導管類型)
 - 術式 (導管與 ablation 機器)
 - 是否研究案? (IIS, CL, DE)

VPC ablation workflow overview

VPC Ablation Workflow - CARTO 3 Modules



Step 1 Pre-case Prep

- 備機
 - PIU
 - Workstation
 - SMA + remote
 - ECG
 - Patch
 - Location pad
- 導管
 - ECG out to recording

Step 2 Setup & Baseline

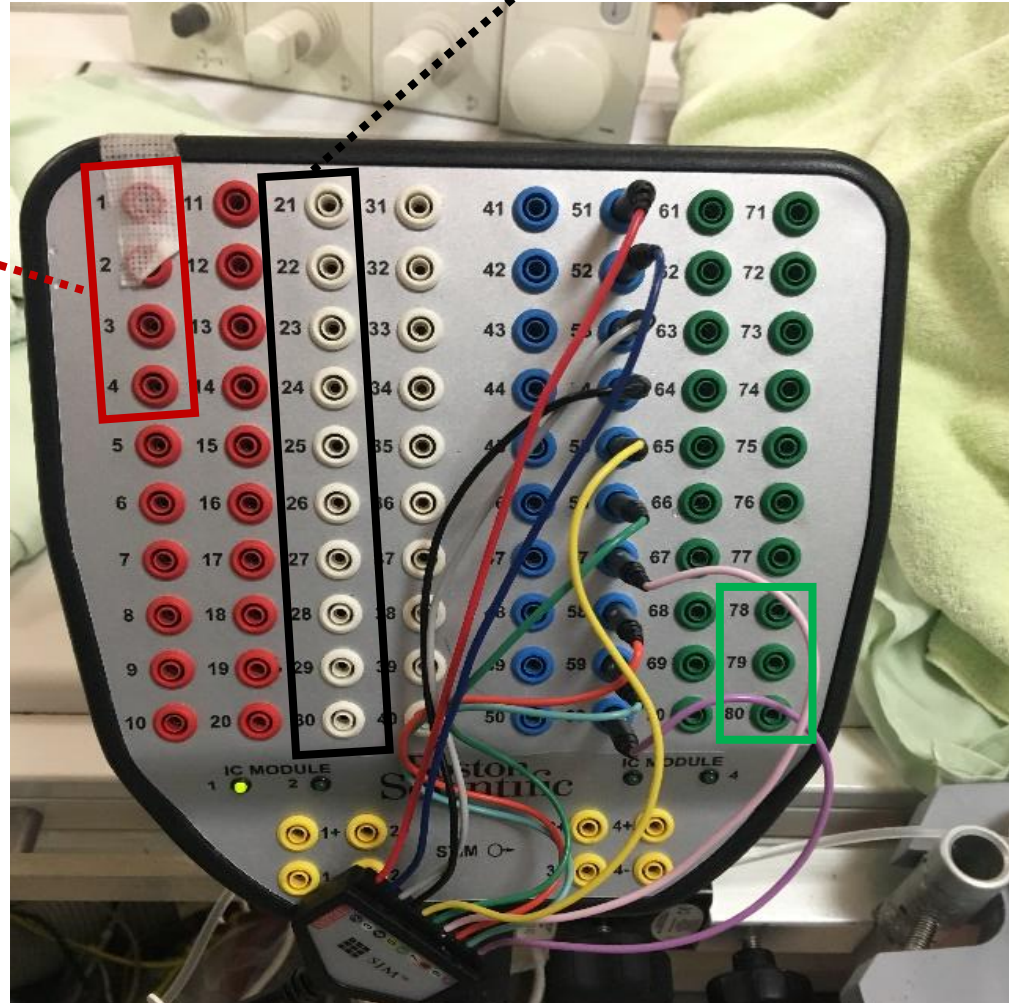
- Catheter 配置
 - 將當日所需的導管放置於control room 外窗邊
- 連接cable 至PIU
 - 將沒有忌諱要先接導管的cable連接至PIU
 - 並將cable 統一掛好於台車上 (方便護理師套無菌套)



IC out (彩虹線) 配置

Pin 21-30
20pA (DECA)

Pin 1-4
MAP (TC)



Step 2 Setup & Baseline

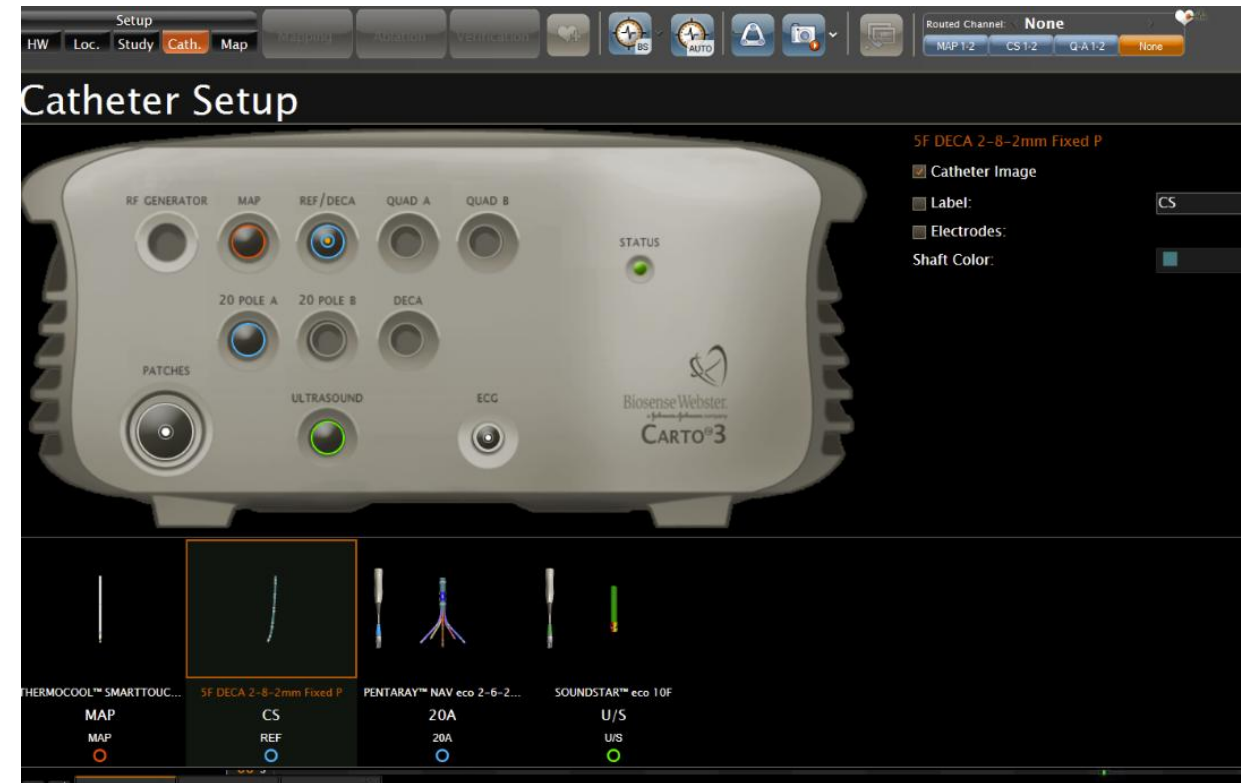
- **CARTO case setup**
 - Study setup: patient name, ID, physician, select template

The screenshot displays the 'Study Setup' window within the 'SIMULATION MODE' of the CARTO3 software. The interface is organized into several sections:

- Navigation Bar:** Includes tabs for 'Study', 'Map', 'Point', 'Catheter', 'ECG', 'Display', 'Imaging', 'Window', 'Tools', and 'Help'. Below these are sub-tabs for 'HW', 'Loc.', 'Study' (which is highlighted), 'Cath.', and 'Map'. There are also buttons for 'Mapplan', 'Ablation', and 'Ventilation', along with icons for 'BS', 'AUTO', and a camera.
- Study Setup Section:**
 - Patient Details:** Contains input fields for 'Last Name:', 'First Name:', 'ID:', and 'Date of Birth:'. The 'Date of Birth' field includes dropdowns for month (currently 'Jan.') and year (currently '1990'). There is a 'Gender:' section with radio buttons for 'Male' (selected) and 'Female'. A 'Patient List...' button is also present.
 - Additional Details:** Includes a 'Comments:' text area, a 'Study Name:' text field, and a 'Physician:' dropdown menu.
- Template Section:** A list of templates for selection: 'Atrial Fibrillation', 'Atrial Flutter', 'Focal Atrial Tachycardia', 'Accessory Pathway', 'AVNRT', 'Ventricular Tachycardia', and 'Other'.
- Connected Machines Section:** Displays the status of connected equipment:
 - 'Recording: No communication'
 - 'Ultrasound: <No Ultrasound>' with a dropdown arrow
 - 'Generators: No Generators'

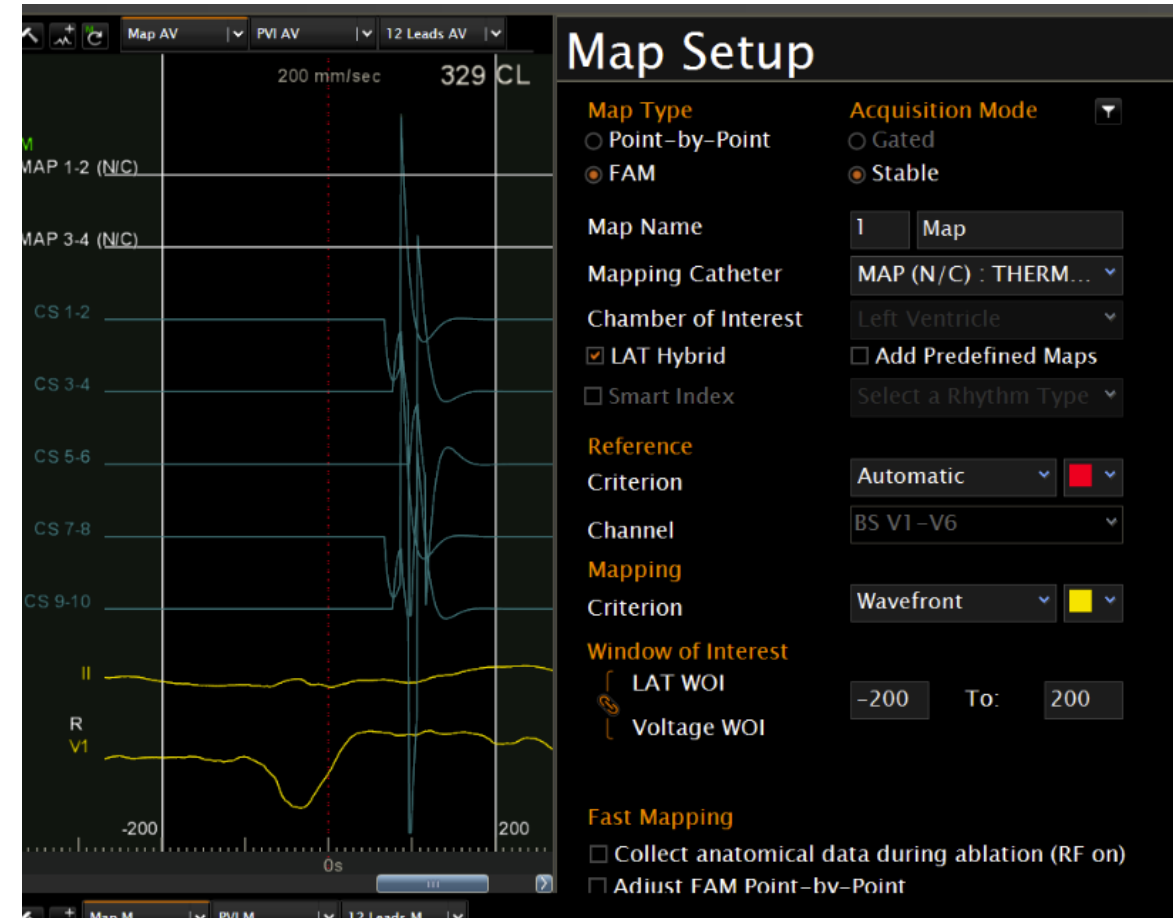
Step 2 Setup & Baseline

- Catheter setup
 - 通常用 DECANAV, THERMOCOOL



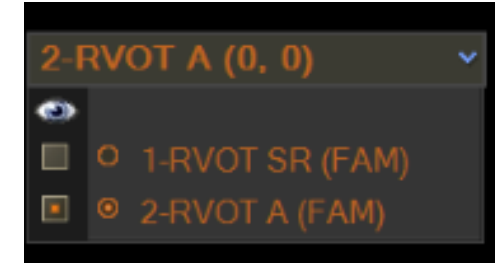
Step 2 Setup & Baseline

- Map setup
 - 先建好幾個可能會用到的圖 (RVOT SR, RVOT A, CS A, CUSP A, RVOT B)
 - Chamber 選擇 Ventricular，勾選LAT hybrid
 - Reference 使用 BS V1-V6 Automatic
 - WOI 預調 -180 to -20



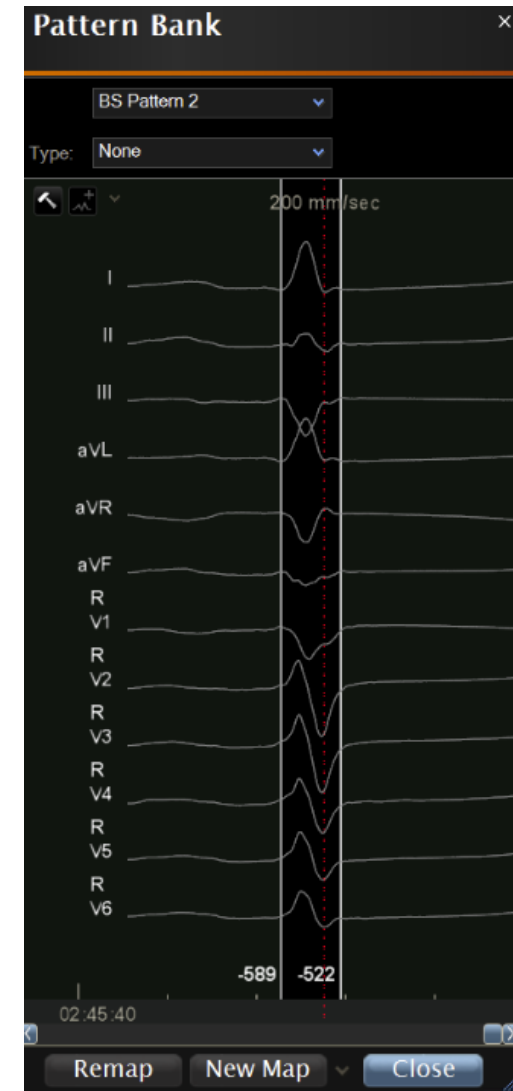
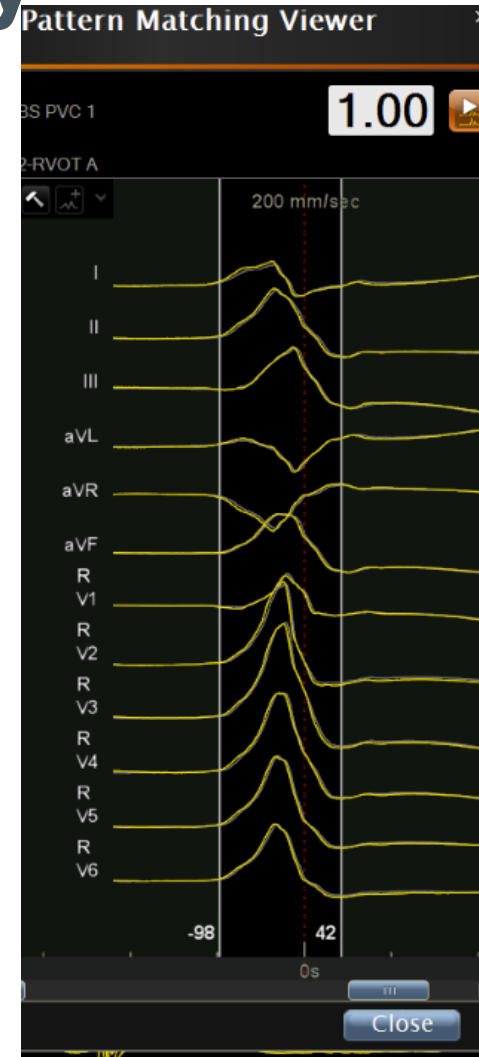
Step 2 Setup & Baseline

- Mapping 頁面
 - 畫面配置: 主畫面左邊 & 副畫面右邊
 - 將預先設定好的Map 進行parallel mapping



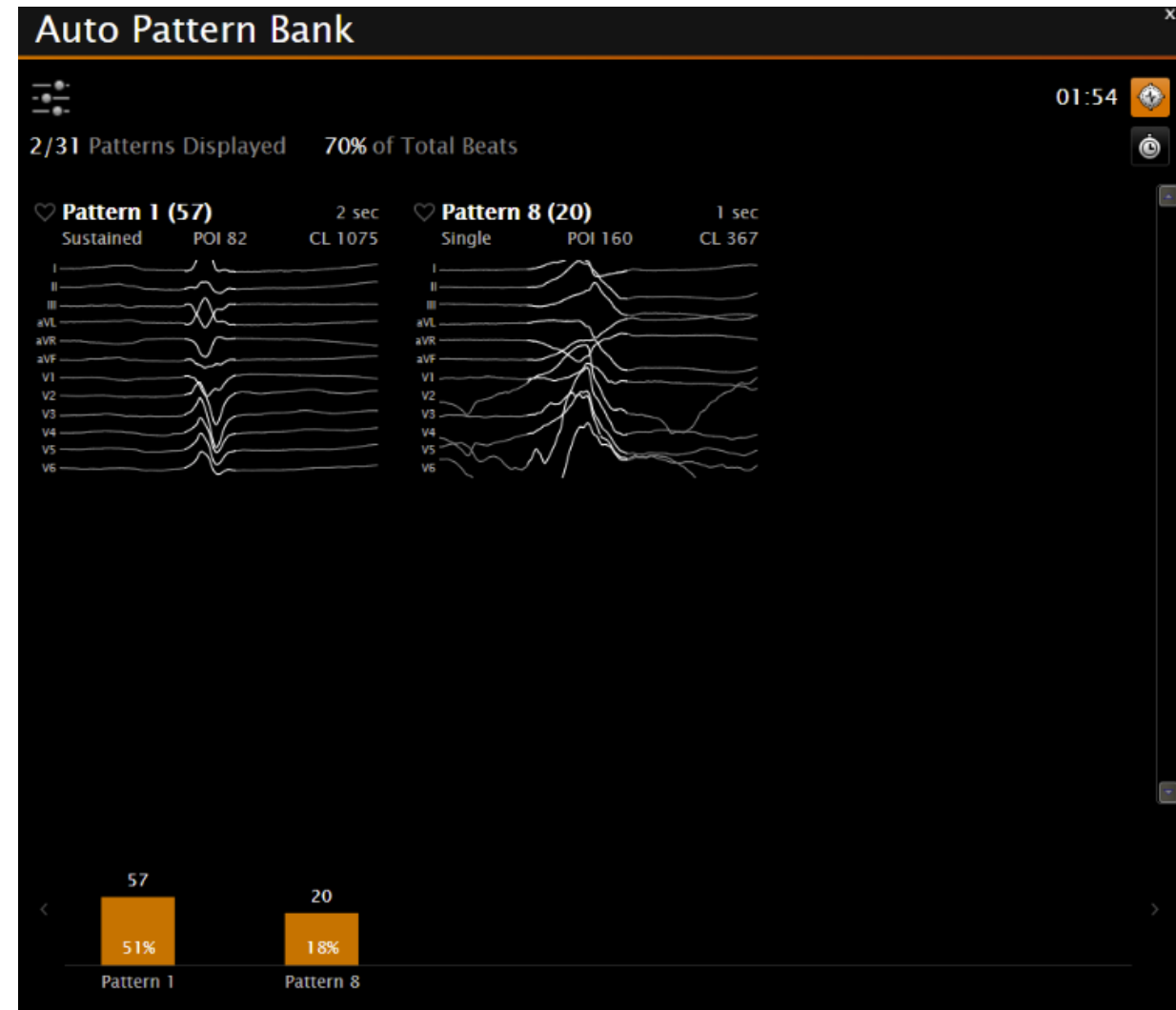
Step 3 Target VPC morphology

- 12 lead ECG 預判
- Template 抓取
 - 選取選單上方的時鐘，使用 pattern matching 並抓取 SR, VPC(可能數個) template
 - 各個Map confidence 設定好pattern matching template, Do not require while pacing 打勾



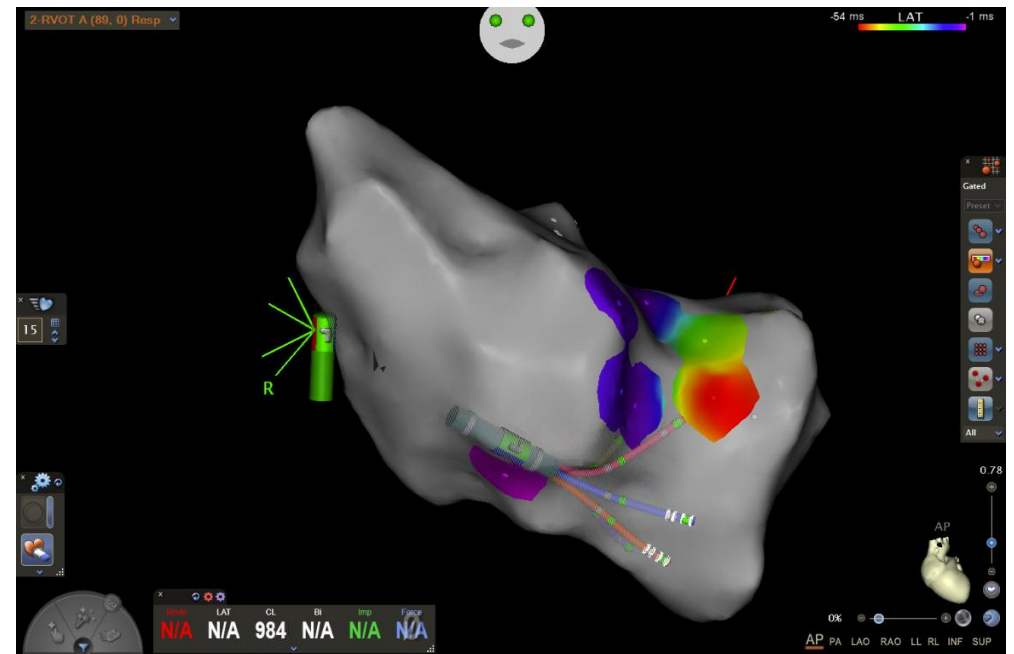
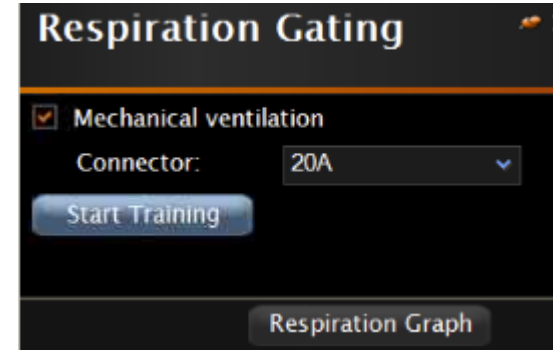
Step 3 Target VPC morphology

- 開啟 Auto pattern bank
 - 收取各 template beat counts



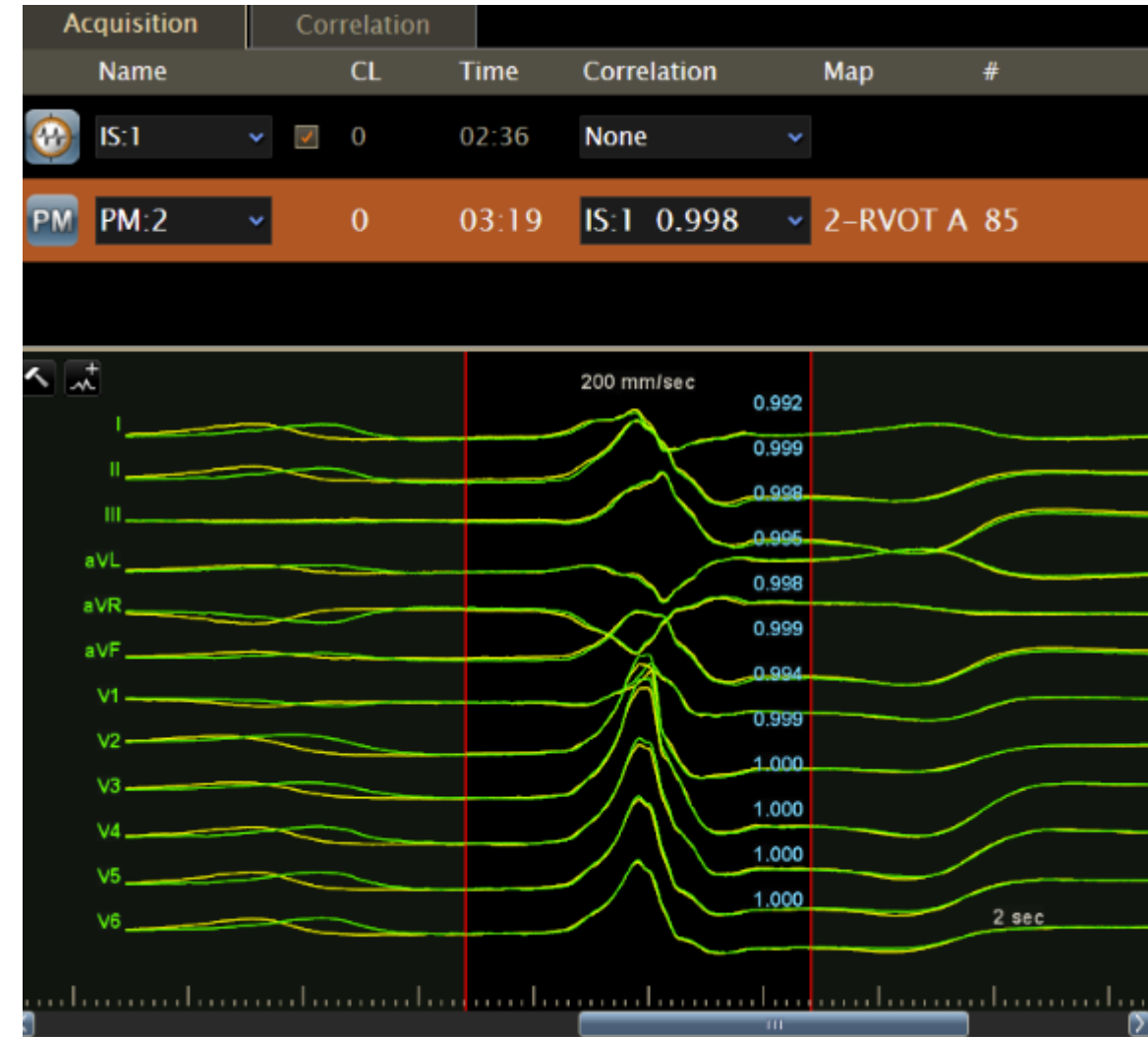
Step 4 Mapping

- 開啟 ACCURESP
 - Enable respiration gating
 - 導管進入後要找地方收呼吸
- 開始用DECANAV 收FAM & EA



Step 4 Mapping

- **PASO 確認相似度**
 - 以臨床 12-lead VT 波形為 template
 - Pace mapping score >90% 視為接近 exit site



Step 5 Ablation

- **Target Selection Checklist**

- 不是 fusion beat
- LAT timing 足夠早
- local EGM 清楚，不是 far-field
- unipolar 呈 QS 或接近 QS
- PASO score 夠高
- catheter contact 穩定
- anatomy 位置合理
- 遠離 His / coronary artery / valve / phrenic nerve
- 若在 cusp / summit，已考慮 coronary distance

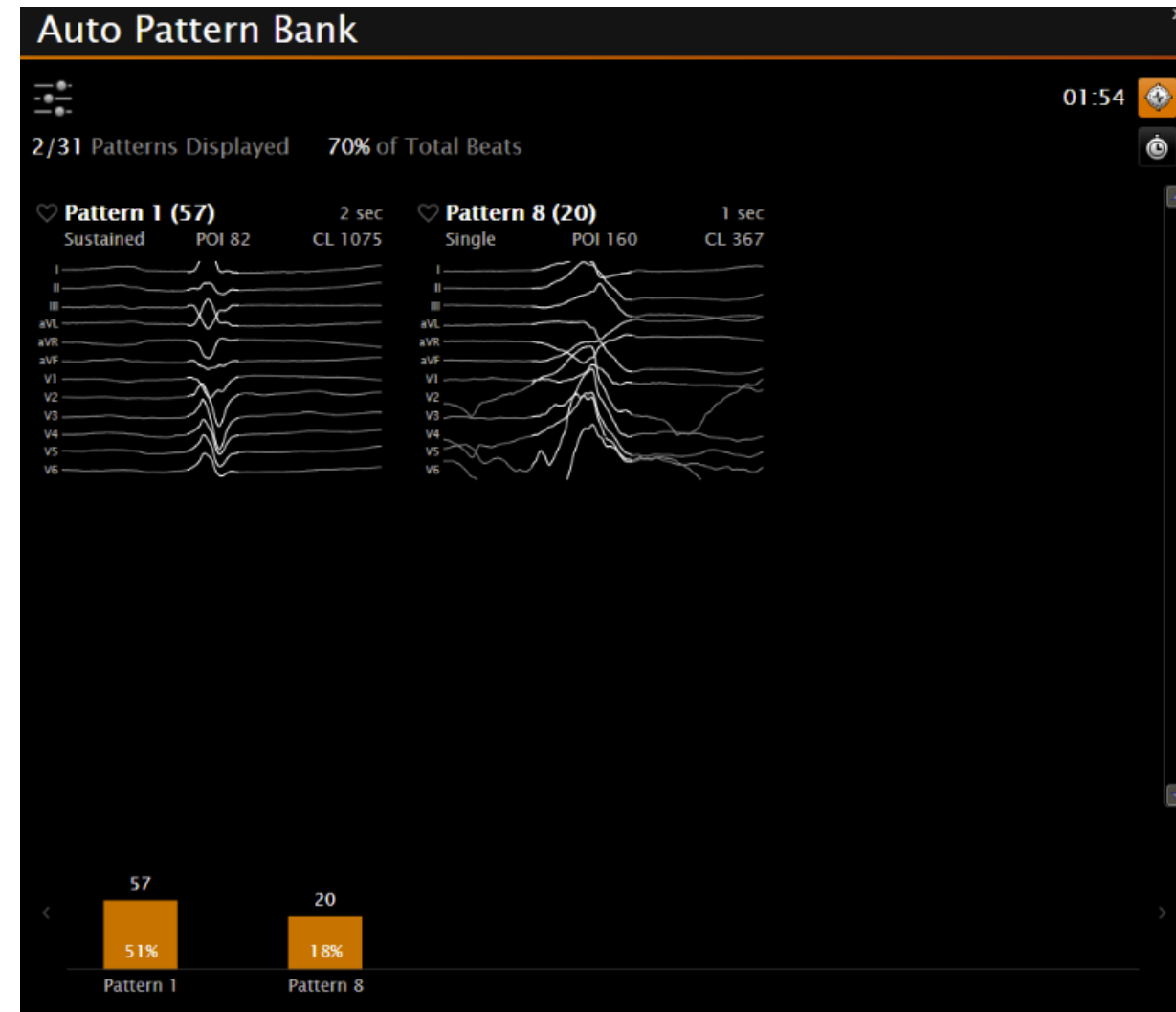
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Step 5 Ablation

- **Post ablation validation**
 - 等待至少 20–30 分鐘觀察 Spontaneous PVC
 - 誘發 (Isoproterenol, Burst pacing)
- **Auto Pattern Bank 確認**





Thank you

